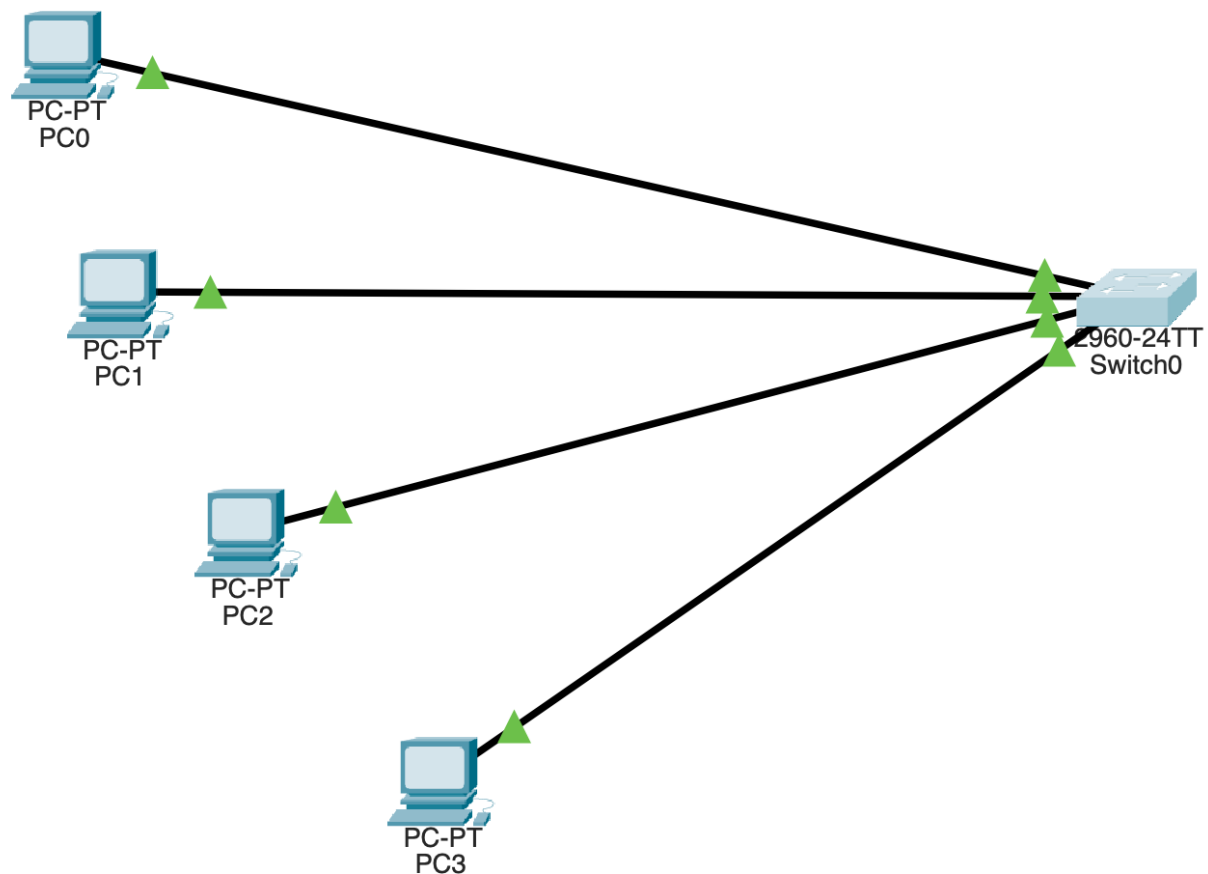
**Lab 12****Course:** Networks System Design**Name:** Do Davin**Student ID:** P20230018**Instructor:** Mr. Kuy Movsun

Part 1: Configuring Port-Based VLANs and Traffic Isolation

- Build the Topology: Connect four PCs to a 2960 Switch using ports Fa0/1 through Fa0/4:



- Create VLANs: Access the Switch CLI and create two VLANs:

```
Switch>enable
Switch#config t
Switch(config)#vlan 10
Switch(config-vlan)#name Engineering
Switch(config-vlan)#vlan 20
```

```
Switch(config-vlan)#name Marketing
Switch(config-vla)#exit
```

- Assign Ports:

```
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
```

- Test Connectivity:

```
C:\>ping 192.168.1.3|

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
```

Part 2: Multi-Switch VLANs and 802.1Q Trunking

- Switch 2:

```
Switch>enable
Switch#config t
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up
```

- Switch 1:

```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk
```

- Inspect the PDU details on the trunk link to find the 802.1Q Tag and the VLAN ID.

PDU Information at Device: Switch0

OSI Model

Inbound PDU Details

Outbound PDU Details

PDU Formats

Ethernet 802.1q

048Bytes

PREAMBLE: 101010..10

DEST ADDR:FFFF.FFFF

SRC ADDR:00D0.978D

TPI

TCl

Type

DATA (VARIABLE LENGTH)

FCS:0x00000000

Arp

0816Bits

HARDWARE TYPE:0x0001

PROTOCOL TYPE:0x0800

HLEN:0x06

PLEN:0x04

OPCODE:0x0001

SOURCE MAC :00D0.978D.B90B

SOURCE IP :192.168.1.1

TARGET MAC:0000.0000.0000

TARGET IP:192.168.1.5

Part 3: Inter-VLAN Routing (Router-on-a-Stick)

• Router:

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#exit
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

Router(config-if)#exit
Router(config)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10,
changed state to up

Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20,
changed state to up

Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router#
```

- Switch 0:

```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed
state to up
```

```
Switch(config-if)#exit
Switch(config)#
%LINK-3-UPDOWN: Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed
state to down

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed
state to up

Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed
state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed
state to up

Switch(config-if)#exit
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#exit
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk
```

- switch 1:

```

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed sta

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEth

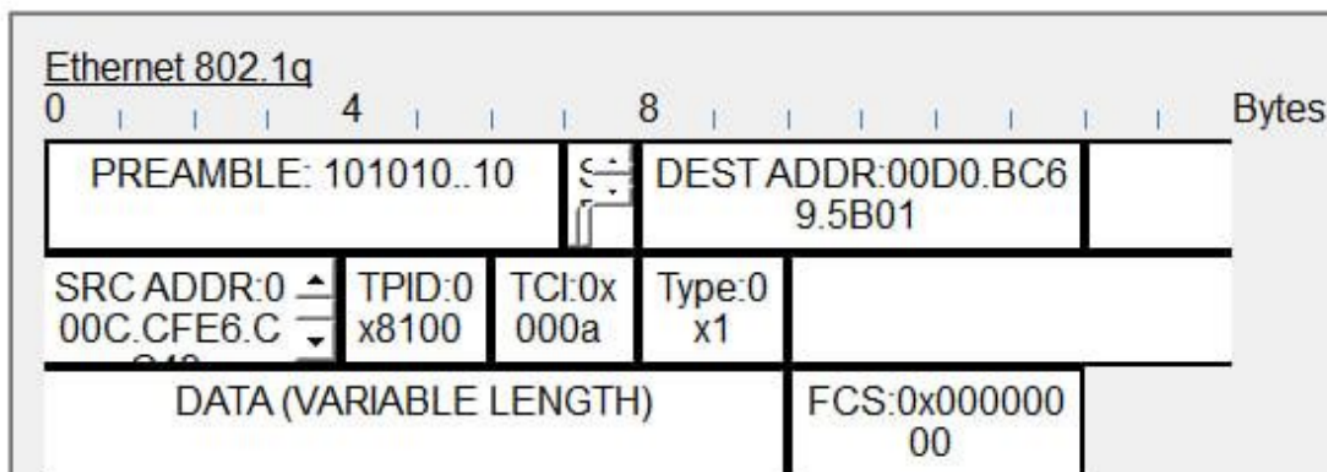
Switch(config-if)#exit
Switch(config)#vlan 10
Switch(config-vlan)#name Engineering
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk

```

PDU Information at Device: Switch0

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats



IP config:

PC0

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.11

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

PC1

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.12

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

PC2

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.20.13

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

PC3

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.20.14

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

PC4

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.15

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

PC5

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.16

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

PC6

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.20.17

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

PC7

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.20.18

Subnet Mask

255.255.255.0

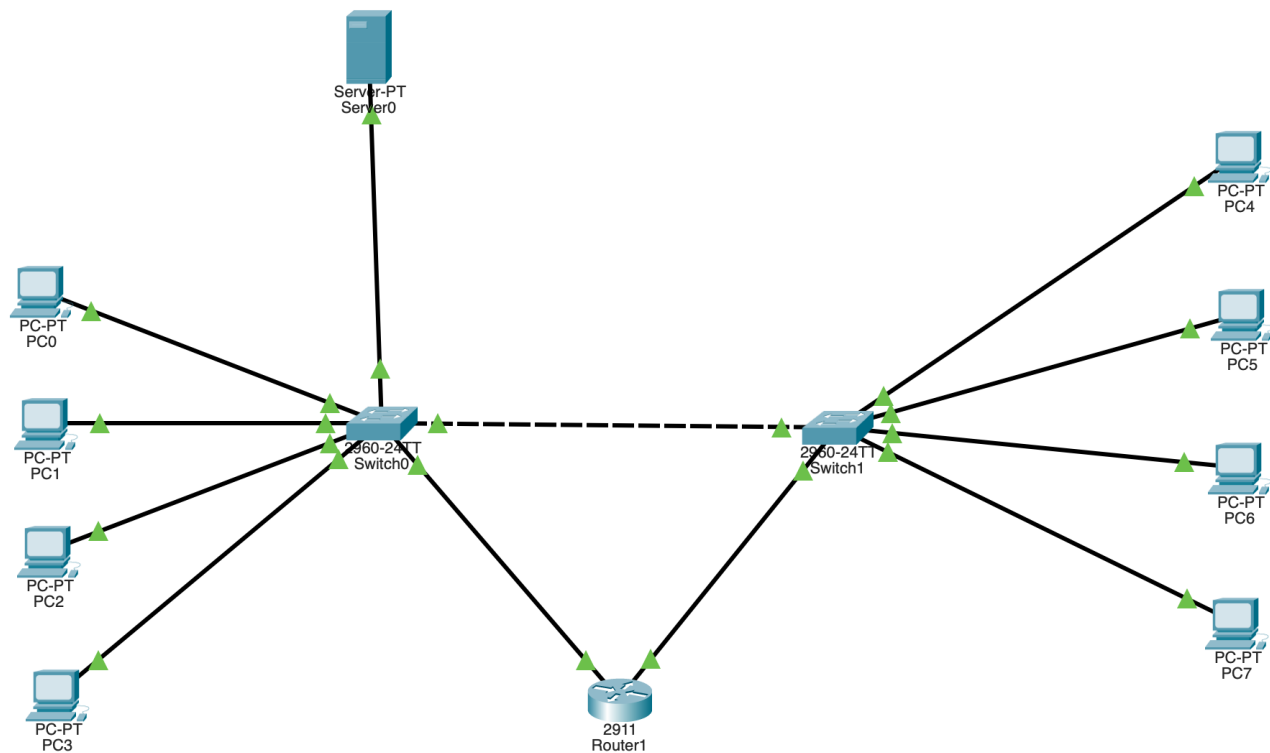
Default Gateway

192.168.10.1

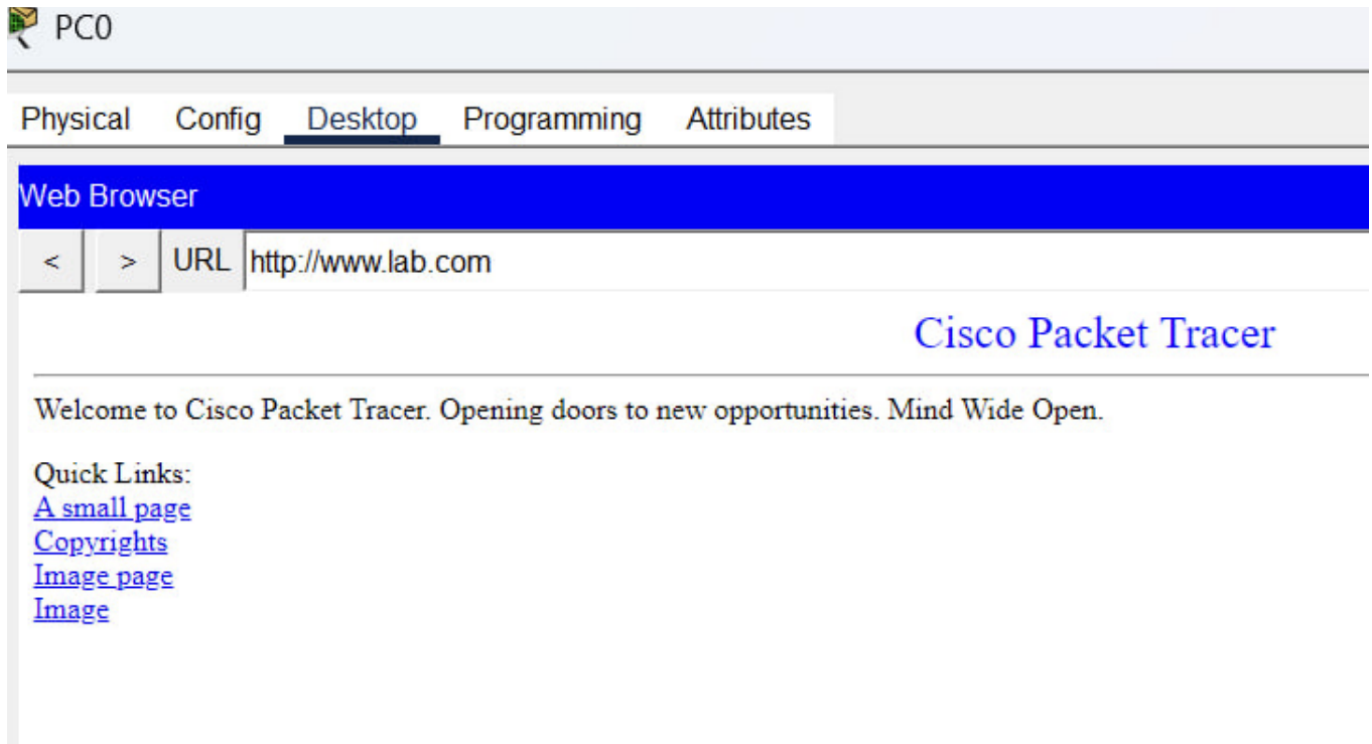
DNS Server

0.0.0.0

Part 4: Synthesis Challenge - "A Day in the Life"



Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	DNS
	0.001	PC0	Switch0	DNS
	0.002	Switch0	Router1	DNS
	0.003	Router1	Switch0	DNS
	0.004	Switch0	Server0	DNS
	0.005	Server0	Switch0	DNS
	0.006	Switch0	Router1	DNS
	0.007	Router1	Switch0	DNS
	0.008	Switch0	PC0	DNS
	0.016	--	PC0	HTTP
	0.017	--	PC0	HTTP
	0.018	PC0	Switch0	HTTP
	0.019	Switch0	Router1	HTTP
	0.020	Router1	Switch0	HTTP
	0.021	Switch0	Server0	HTTP
	0.022	Server0	Switch0	HTTP
	0.023	Switch0	Router1	HTTP
	0.024	Router1	Switch0	HTTP
	0.025	Switch0	PC0	HTTP



1. Router-on-Stick & Trunk Mode:

A single physical connection can be used to transport traffic from multiple VLANs to a router. By using trunk mode with 802.1Q tagging, VLAN information is preserved in each frame, allowing the router to identify the source VLAN and perform inter-VLAN routing correctly.

2. 12-bit VLAN ID Purpose:

The 12-bit VLAN ID field in an 802.1Q frame is used to uniquely label VLANs in the range of 1 to 4094. This identifier allows switches and routers to recognize which VLAN a frame belongs to, ensuring proper traffic separation and forwarding.

3. VLANs & Broadcast Domain Reduction:

VLANs divide a network into multiple broadcast domains. Broadcast traffic is restricted to the VLAN in which it originates and is not forwarded to other VLANs, which helps reduce unnecessary broadcast traffic and improves overall network efficiency.